

1 Solution — GIAM 7.1

Problem 7

1.1 Problem

How many “words” are there of length 4 from the Cryptographer’s alphabet, with *repeated letters allowed*, in which the letters appear in *non-decreasing* order alphabetically?

1.2 Setup

The alphabet is $\Sigma = \{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e}, \mathbf{f}\}$ with its natural total order $\mathbf{a} < \mathbf{b} < \mathbf{c} < \mathbf{d} < \mathbf{e} < \mathbf{f}$.

A *valid word* is a tuple $(x_1, x_2, x_3, x_4) \in \Sigma^4$ satisfying the *weak* inequality

$$x_1 \leq x_2 \leq x_3 \leq x_4.$$

The crucial difference from Problem 6 is that equality is now permitted: words such as “aaaa”, “aabc”, and “eeff” are all valid.

1.3 Count — Multisets in Disguise

Define the map

$$\Phi : \{\text{valid words}\} \longrightarrow \{\text{size-4 multisets of } \Sigma\}, \quad \Phi(x_1, x_2, x_3, x_4) = \{\{x_1, x_2, x_3, x_4\}\},$$

where $\{\{ \cdot \}\}$ denotes a multiset (an unordered collection with multiplicities). Then Φ is a *bijection*:

- **Well-defined.** $\Phi(\cdot)$ is by construction a multiset of size 4 over Σ .
- **Surjective.** Any size-4 multiset of Σ admits *exactly one* non-decreasing arrangement (sort its elements in alphabetical order), and that arrangement is a valid word with the given Φ -image.
- **Injective.** Two valid words with the same underlying multiset must each list it in the unique non-decreasing order, hence be equal.

So counting valid words is the same as counting size-4 multisets of a 6-set. This is the canonical *stars and bars* problem.

1.3.1 Stars and Bars

Encode a size-4 multiset of Σ as a row of 4 identical *stars* and 5 identical *bars*: the bars carve the row into 6 bins (one per letter **a, b, ..., f** in order), and the number of stars in each bin records that letter's multiplicity. For example, the multiset corresponding to the word "**aabe**" — namely $\{\{\mathbf{a}, \mathbf{a}, \mathbf{b}, \mathbf{e}\}\}$ with multiplicities $(2, 1, 0, 0, 1, 0)$ — is encoded by the row

$$\star\star \mid \star \mid \mid \mid \star \mid ,$$

a sequence of 9 tokens.

The number of such sequences is the number of ways to choose the 4 positions occupied by stars among the 9 total positions:

$$\#\text{valid words} = \binom{6+4-1}{4} = \binom{9}{4} = \boxed{126}.$$

By the symmetry identity proved in Problem 4, this also equals $\binom{9}{5} = 126$ — i.e., choosing the 5 bar positions instead.

A direct arithmetic check:

$$\binom{9}{4} = \frac{9 \cdot 8 \cdot 7 \cdot 6}{4!} = \frac{3,024}{24} = 126.$$

1.4 Enumeration — Hockey Stick by First Letter

After fixing x_1 , the tail (x_2, x_3, x_4) is a non-decreasing length-3 word from $\{\ell \in \Sigma : \ell \geq x_1\}$ — equivalently, a size-3 multiset of that smaller alphabet. If m letters remain (i.e., **a** leaves $m = 6$, **b** leaves $m = 5$, etc.), the tail count is $\binom{m+2}{3}$:

First letter x_1	m remaining	Tail count $\binom{m+2}{3}$
a	6	$\binom{8}{3} = 56$

b	5	$\binom{7}{3} = 35$
c	4	$\binom{6}{3} = 20$
d	3	$\binom{5}{3} = 10$
e	2	$\binom{4}{3} = 4$
f	1	$\binom{3}{3} = 1$
Total	—	$56 + 35 + 20 + 10 + 4 + 1 = 126$

Table 1.1

This decomposition is the *hockey-stick identity*

$$\binom{9}{4} = \sum_{i=3}^8 \binom{i}{3} = 1 + 4 + 10 + 20 + 35 + 56 = 126,$$

confirming the count.

1.5 Remark — Shift Bijection to Problem 6

A second classical proof relates this problem directly to Problem 6. Encode $\mathbf{a} = 1, \mathbf{b} = 2, \dots, \mathbf{f} = 6$, so $x_i \in \{1, 2, \dots, 6\}$, and define

$$y_i := x_i + (i - 1), \quad i = 1, 2, 3, 4.$$

Then $y_i \in \{1, 2, \dots, 6 + 3\} = \{1, \dots, 9\}$, and the weak inequality $x_i \leq x_{i+1}$ becomes the strict inequality

$$y_{i+1} - y_i = (x_{i+1} - x_i) + 1 \geq 1 \implies y_i < y_{i+1}.$$

So non-decreasing length-4 words from a 6-alphabet correspond bijectively to *strictly* increasing length-4 words from a 9-alphabet — exactly the setup of Problem 6 scaled up. The count is therefore

$$\binom{9}{4} = 126,$$

agreeing with the stars-and-bars answer.

1.6 Remark — Hierarchy of Length-4 Counts

Filling out the table from Problem 6 with this problem in place:

Constraint on the length-4 word	Count
any letters, any order (Problem 5)	$6^4 = 1,296$
<i>distinct</i> letters, any order	$6 \cdot 5 \cdot 4 \cdot 3 = 360$
any letters, <i>non-decreasing</i> (this problem)	$\binom{9}{4} = 126$
<i>distinct</i> letters, <i>increasing</i> (Problem 6)	$\binom{6}{4} = 15$

Table 1.2

The bottom two rows are *unordered* counts — each represents a single canonical arrangement (sorted) of a multiset or subset respectively. The connection across rows is the standard pair of identities

$$\binom{n}{k} = \frac{P(n, k)}{k!}, \quad \binom{n+k-1}{k} = \text{number of size-}k \text{ multisets of an } n\text{-set.}$$

The subset coefficient $\binom{6}{4} = 15$ counts strict, distinct arrangements; the multiset coefficient $\binom{9}{4} = 126$ counts weak, repetition-allowed arrangements. Their ratio $126/15 = 8.4$ has no clean combinatorial meaning (since multiset arrangements have varying multiplicities), but the *count* of multisets minus subsets — namely $126 - 15 = 111$ — gives precisely the number of size-4 multisets of Σ in which at least one letter repeats.